Chapter 14

Essentials of Pharmacoepidemiology

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Learning Objectives:

Objective 14.1 Outline pharmacoepidemiology and its methodological terminologies.
Objective 14.2 Explain study designs used in pharmacoepidemiology research.
Objective 14.3 Discuss how pharmacoepidemiology influences health improvement.

OBJECTIVE 14.1. OUTLINE PHARMACOEPIDEMIOLOGY AND ITS METHODOLOGICAL TERMINOLOGIES

The classical definition of pharmacoepidemiology is the study of the use and the effects of drugs in large numbers of people.[1] It is a combination of two important components: pharmaco, related to pharmacology or drugs, and epidemiology, the study of determinants and distribution of disease in a population (Fig. 14.1). Therefore, pharmacoepidemiology is a specialized category of the general study of epidemiology where medications are the primary exposure in a population and the determinants of the distribution of the outcome of interest in a large population are investigated. What is also necessary for the study of pharmacoepidemiology is the understanding of clinical practice and the behaviors of clinicians in relation to prescribing and decision making.

In the drug approval process, there are inherent limitations. Randomized clinical trials are used to understand the efficacy of the drug (can the drug work in the body in experimental conditions?) and effectiveness (does the drug provide a desired effect in the real world?), but may be limited by sample size and follow-up time. To clearly understand the drug effects, whether wanted or unwanted, a variety of patient characteristics in a large population are to be observed. It will identify uncommon effects that may not occur immediately, but later in time (e.g., causing cancer). Therefore, a study of the medication used in a *real-world* population, including women, young people, and elderly patients, allows the researcher to follow a patient over several years to identify benefits and adverse drug reactions in a variety of patient characteristics. These studies are conducted usually after the drug has entered clinical practice and are called postmarket surveillance. Pharmacoepidemiological studies not only identify the determinants that may cause bad outcomes but also identify those patients who would most benefit from the medication under study.

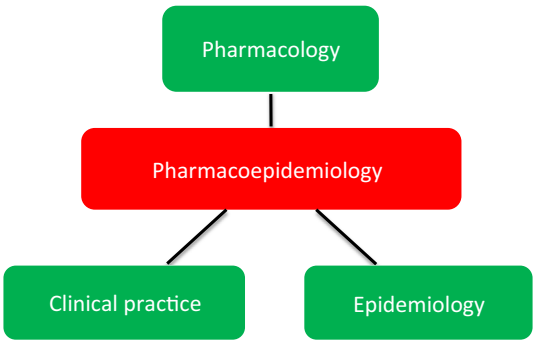


FIGURE 14.1 The components of pharmacoepidemiology.

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Some examples of where pharmacoepidemiological studies may be used include identifying or providing the evidence for best practice that can be used further in evidence-based practice. Drug safety studies may require large populations of patients taking medication in a variety of patient characteristics with varying degrees of disease severity to identify those at most risk of adverse drug reactions. If a drug has a rare adverse drug event, it may need time to be identified and a large number of people to find it.

As a hospital pharmacist, the results and evidence from robust pharmacoepidemiological studies can provide the evidence for therapeutics and formulary choices to plan the best care possible based on research. The evidence from research may show that not everyone will benefit from a particular drug. This may prevent the use of and costs associated with expensive therapies given to patients that may provide suboptimal benefit from the treatment. However, the evidence will also identify the patients that may get effective treatment and perhaps a cure.

In community pharmacy, the evidence from pharmacoepidemiology studies can help provide the best practice for patients who need it the most. The results will help in decision making for clinically effective and cost-effective therapies to use in a particular disease management scheme or evidence to change to a different class of drug.

Link to Pharmacoeconomic Studies

In pharmacoepidemiology, drugs are the primary exposure agent of interest. Drugs cost money to supply to the pharmacy and cost the patient/insurance/government money to buy or provide. Therefore, a cost can be identified. The evidence from pharmacoepidemiological research can, therefore, feed into pharmacoeconomic research providing preferences and effectiveness for such study designs as cost-effective analysis or other analyses.

Methodological Terminology

The following are terms that are often used in pharmacoepidemiology and need to be understood so that the correct interpretation of the data and study design are performed.

Confounding

A confounder is a variable that is associated or has a relationship with both the exposure and the outcome of interest. Fig. 14.2 shows an example of a possible confounder found in the relationship between alcohol use and lung cancer. By itself, this relationship does not make clinical sense; how can drinking alcohol cause cancer? However, when the confounding variable smoking is introduced, the relationship makes more sense; an individual who drinks alcohol is more likely to smoke, and smoking could cause cancer. The variable is related to alcohol use and lung cancer and is part of the causal pathway. Confounding can be controlled for in a study by either matching the confounding variable to ensure equal proportions of the confounder are found in each group or by statistical methods (i.e., multiple variable regression analysis).

Bias

Bias is defined as a difference between groups that favors one group over another.[2] This cannot be controlled by statistical analysis and is a fact of the data. For example, a small group of cancer patients exposed to a particular agent is coded for severity of disease more accurately than another group in a study without the exposure. There is a difference in the coding accuracy between the groups. This would introduce bias to the study of disease severity.

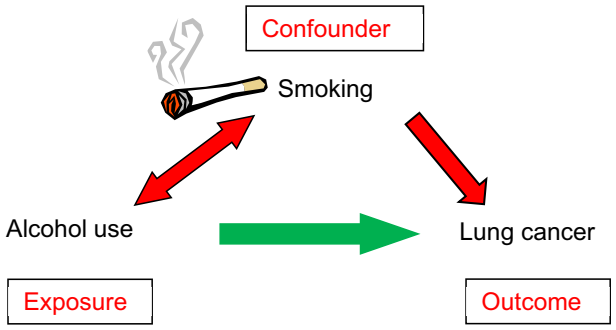


FIGURE 14.2 Illustration showing confounding and the relationship between exposure and outcome.

Incidence

Incidence of disease is defined as the number of newly diagnosed cases of the disease in a population divided by the total population at risk of the disease.[2] New cases of the disease have not had a prior diagnosis of the disease. The calculation of incidence is as follows:

$$\mathrm{I} = \frac{\text{Number of people with a new diagnosis of disease}}{\text{Number of people at risk}}$$

Point Prevalence

Prevalence of disease is defined as the number of people with an existing disease (both old and new cases) in a population divided by the total population at risk of the disease.[2] Some of the people could have had the disease for a long time and some may have been diagnosed recently, but all are included. It is the proportion of the population with the disease. This is a useful calculation to quantify the burden of disease in a population. Point prevalence is the number of people with the disease of interest at a specific point in time divided by the number of people in the population at the specific point in time. Point prevalence is calculated using the following equation:

$$\text{Point prevalence} = \frac{\text{Number of people with new and existing disease}}{\text{Number of people at risk at a specific point in time}}$$

Matching

Matching is when one group is made similar to the comparator group by making pairwise matches on a small number of known risk factors. The risk factors are known confounders in the relationship, and the researcher would like to ensure the study is measuring the true relationship between exposure and outcome without being interrupted by other possible risk factors. Matching is usually performed using a small number of variables to ensure that there are sufficient numbers in each group. The more risk factors used in matching, the smaller the number of pairwise matches that can be found in each comparison group. For example, age and sex of the patient are known risk factors for myocardial infarction (MI). As the male subject grows older, they are at increased risk. Matching each individual in the groups by sex and age ensures the groups have a similar distribution of these variables and a more precise final measure of association between the exposure of interest and the outcome. If smoking habits and other lifestyle risk factors are to be matched on, the comparator groups will naturally become smaller with increased number of matching variables.

Drug Utilization

Drug utilization focuses on the various medical, social, and economic aspects of drug use. Medical consequences include the risks and benefits of drug therapy, whereas social aspects can be related to inappropriate use. Economic issues deal with the cost of drugs and treatment for patients and society.[3] This is different to pharmacoepidemiology because drug utilization does not necessarily have to link to a health outcome event. Pharmacoepidemiology always studies both exposure and outcome. Drug utilization is more encompassing and includes social/behavioral factors, which may affect why patients use medications; this is not usually the focus of pharmacoepidemiological research.[3]

Adherence

Adherence and compliance are often used interchangeably. Compliance is the patient's ability to take a medication as prescribed, which sounds like a command to take a medication.[3] If a patient does not take medication as prescribed, they may be thought to have bad medication-taking behavior.[3] Adherence is more complex and multifactorial than compliance. Adherence has issues of patient behavior, patient–doctor relationships, and pharmacist–patient relationships that have to be accounted for or at least identified to fully understand why a patient has decided to take or not to take their medication as directed on the label. Adherence to medication should be taken into account in the analysis of the data because it does relate to the dose of the exposure of interest. Adherence is usually measured as a percentage of time in possession of a medication, assuming the patient is taking the medication the whole time and as directed. Another term that is related to adherence is persistence. This term is used to explain the extent to which the treatment is taken for the recommended duration of therapy.[3] Persistence is used more often in long-term and chronic conditions where taking medication regularly for long periods of time is important to the treatment of the chronic disease.

Disease Severity

Disease severity is another measure that may affect the measure of association in a pharmacoepidemiological study. The patient may have tried many medications as the disease has progressed and is using a new medication as the last hope of treatment. In a study, the new medication may be associated with a bad outcome, not because it is the cause, but because the severity of the disease has caused the patient to fail. For example, a patient with painful arthritis has tried different NSAIDs available and is finally switched to a new analgesic agent but fails (has a joint replacement). The drug should not be associated with the failure of treatment because it was another chance taken before attempting a major invasive intervention.

OBJECTIVE 14.2. EXPLAIN STUDY DESIGNS USED IN PHARMACOEPIDEMIOLOGY RESEARCH

Ecological/Cross-Sectional Studies

For some research questions, the only data available are aggregated at group level instead of the individual patient level. For example, data on cough and cold preparations bought over-the-counter (OTC) in a town's two pharmacies. Here, the total number of packages sold by the pharmacies each day is recorded, but no data are collected on the individual purchases by patients/customers. Analysis of aggregated data of daily purchases can inform decision-makers on what types of products are sold or required and may inform public health if there is a possible outbreak of upper respiratory infection in the area. The data do not necessarily relate it to a particular outcome of interest but can provide information on the use of medications in a particular population. The results of this type of study design can be compared with other areas to investigate if there is a difference in the rate of daily purchases, which further inform area public health officials of the geography of the outbreak. Studies using aggregated data are called *ecological studies*.

There are limitations to ecological study design, namely confounding. Confounding arises because there is a lack of individual patient information. When analyzing aggregated data, information may be lost that could provide an explanation on the use of cough and cold products. For example, there may be a school with young children attending increasing the risk of spreading a cold to a large population (students and parents) or an area of elderly less prone to flu because of vaccination programs. This could lead to an *ecological fallacy* where the researcher makes conclusions at an individual level when only aggregate data are used.[4]

The measures for aggregated data are rates. This is a simple calculation of the number of people that are affected by the total population that are at risk. For example, using the above situation, there were 130 purchases of OTC cough and cold products in 1 day from both pharmacies. The town has a total population of 5145 people. Therefore, the rate of OTC cough and cold purchases is $130/5145 = 0.025$ or 25 purchases per 1000 people.

Case–Control Studies

Case–control study designs are used widely in pharmacoepidemiology. They are particularly useful when the outcome of interest is relatively rare. For example, to study the relationship between drug exposure and a specific type of cancer, a case–control design could be used easily.[5] Case–control studies begin with the identification of a sample of patients with the outcome of interest (e.g., death, cancer, or MI); these are the cases. The controls are identified from the same study population that does not have the outcome of interest (e.g., not dead, do not have cancer, and do not have an MI). Within each sample, the researcher then looks back in the medical record to determine the exposure status to a drug of interest (Fig. 14.3). This type of study is retrospective in time because while the data are collected, the outcome and exposure have already happened. The researcher can now calculate the proportion of the case sample and the control sample that was exposed to the drug of interest in the past. We can compare the proportions and calculate the *odds ratio* (OR) (odds of exposure in cases vs. odds of exposure in the controls).[2]

This study design is efficient because the research is not waiting for the outcome to develop after drug exposure. The data collection starts with the outcome and can easily look back in time to see if they were exposed. With this design, the researcher can examine multiple exposures or risk factors of interest for a given outcome, as long as the exposure happened before the outcome. To do this, the case is assigned an index date, which is the date the outcome occurred. Controls (without the outcome of interest) are also given the same index data as their matched case (if matched on a particular variable, e.g., age) or randomly selected case. Exposure status is determined retrospectively from the index date.

Because the study design uses a sample of the study population to allocate the cases and controls, the incidence of disease in the study population cannot be calculated. It is difficult to identify all those that are at risk in the study population. However, the calculated OR could be viewed as a valid estimate of the relative risk (RR) in certain circumstances. These circumstances include the following: (1) cases are representative of the source population; (2) controls should be

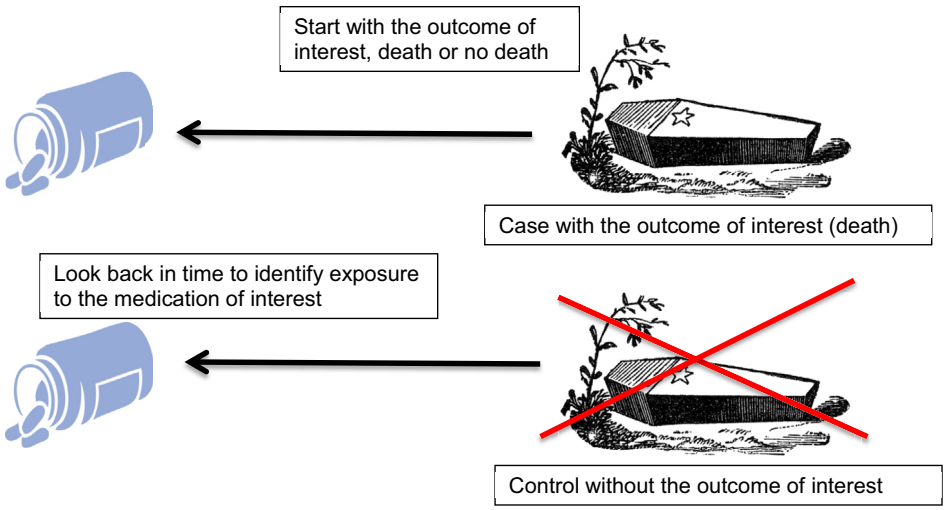


FIGURE 14.3 Case–control study design and timeline for data collection.

selected from the same population; and (3) the controls should be sampled regardless of their exposure status.[2] One way of ensuring that these circumstances are fulfilled is to ensure that eligibility criteria are applied to both cases and controls in the study.

There are some limitations to this study design the researcher should consider. First, the study design is susceptible to bias and confounding, which can have an effect on the final results and interpretation of the study. *Selection bias* may occur if the cases and controls are systematically different from those that are not included in the study. *Information bias* may occur when data are more complete for cases compared with controls, leading to misclassification of disease or outcome or measurement error. *Recall bias* may occur if exposure status is remembered more accurately in cases because the outcome is worse. For example, a life-threatening adverse drug reaction may have exposure to a drug more accurately remembered than those without an adverse drug reaction.[6]

Confounding is a limitation that is found in this study design but can be controlled by matching or statistical analyses. Confounding happens when a variable is related to both the outcome and the exposure. Matching the cases and controls on a confounding variable will ensure that the variable is the same in both groups, decreasing the likelihood of effects on the final results. Statistically controlling the confounding "adjusts" the proportion of the confounder in each group to ensure that they are the same in each group before calculating the final adjusted OR. This is performed using multivariable regression analysis.

Measure of Association for a Case–Control Study

The measure of association for a case–control study is the OR. The calculation of the OR is relatively easy if some simple rules are established and the researcher sets up their results appropriately to identify the exposure and outcome. Table 14.1 shows the calculation of the OR. Interpreting the OR can be remembered by using the result to answer cell "A." For example, there is an increased/decreased risk of the outcome if exposed to the medication.

Case Study: Case–Control Study Design

Prescription-acquired acetaminophen use and the risk of asthma in adults: a case–control study. Kelkar M, Cleves MA, Foster HR, Hogan WR, James, LP, Martin BC. *The Annals of Pharmacotherapy*, 2012;46:1598–1608.

This study investigates the link between acetaminophen (paracetamol) use and the development of asthma. The authors provide evidence that acetaminophen can cause lung damage, which could lead to asthma. This case–control study design started with the first diagnosis of asthma and looked back to observe acetaminophen use. They used many confounders in the analysis to find the adjusted association. Cases were found to be more ill and used acetaminophen on a chronic basis, whereas controls did not. Some confounders were not found in the data (smoking). The study found that chronic use of high doses of acetaminophen is associated with asthma (OR 1.70 95% CI 1.63–1.98). However, most of the preparations also contain codeine, which may cause respiratory depression and should be taken into consideration when discussing the study results.

TABLE 14.1 Calculation for the Odds Ratio

	Outcome of Interest	
	Yes (Cases)	No (Controls)
Exposed to medication	A	B
Not exposed to medication	C	D

A, exposed and has the outcome; *B*, exposed and does not have the outcome; *C*, not exposed and has the outcome; *D*, not exposed and no outcome; *OR*, AD/BC.

Cohort Studies

Cohort studies are another popular study design used in pharmacoepidemiology. This type of study is often used when a randomized control trial cannot be undertaken because of requirements for exposure (pregnancy or elderly patients) or the exposure is either impossible or unethical (effects of smoking or taking illegal drugs). The main purpose of the cohort study is to estimate the risk or rate of an outcome among a cohort of individuals.

A cohort study initially starts by identifying a large group of people that are free of the outcome of interest. This is in contrast to the case−control study where the study starts with the identification of people with the outcome of interest. The cohort of disease-free people is now assigned to either exposed or not exposed (e.g., smoking or not smoking) and followed forward in time until the outcome of interest occurs (Fig. 14.4). This may take a long time, so cohort studies may take years or generations until an outcome is identified. The cohort study is prospective in design because it follows people forward in time gathering information along the way. There are particular scenarios where a researcher may say their study is a retrospective cohort study, meaning that samples are collected at the beginning of the study but analyzed when the outcome of interest is identified in the future. This saves a lot of time and money if the analysis is laborious and costly.

For example, the WHO MONICA Project is a large cohort of 10 million people aged 35−64 from 26 countries of Europe, North America, and Western Pacific. Subjects were followed forward in time to a fatal or nonfatal cardiovascular event or stroke. Physical examinations and surveys were performed at regular intervals under a standardized protocol. This study has provided information on risk factors contributing to MI and medications used to control blood pressure and their effects on the mortality rate of acute MI.[7]

Measure of Association for a Cohort Study

Because the cohort study design is prospective, an accurate record of exposure can be recorded, including adherence to any medication and other health effects that occur over time (e.g., flu exposure). The temporal relationship between the exposure and the outcome can also be investigated, which can be important when establishing a causal relationship.[2] A major advantage of cohort studies is that the entire population is at risk because the study population started the study outcome free. Therefore, an accurate *incidence* of the outcome can be calculated for the exposed (I_e) and unexposed (I_0) populations (Fig. 14.5). The incidence is used in the measure of association termed the *relative risk* (RR). The calculation of the RR starts with a similar table as the case−control study, but different information is used because we are interested in the relationship between the incidence of the outcome in the exposed and unexposed.

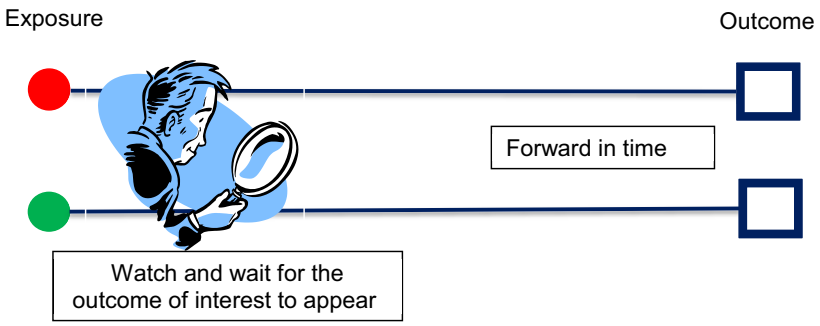


FIGURE 14.4 The temporal relationship of exposure and outcome in a cohort study design.

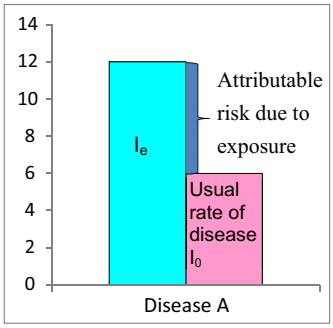


FIGURE 14.5 Explanation of attributable risk in relation to the incidence of disease in the unexposed (I_0).

Knowing the incidence of the outcome of the exposed and not exposed, we can also perform other useful information that can be used in the clinical setting. Attributable risk is the quantification of risk that can be attributed to the exposure alone. It is calculated by taking the absolute value (only positive values) of the difference between the incidences in the exposed and unexposed. The numbers needed to treat or harm can be calculated by taking the inverse value of the AR (1/AR), which will provide the number of individuals who have to receive the treatment for one of them to benefit from the treatment over a specific time.

Case Study: Framingham Heart Study

In 1948, 5209 adult residents of the town of Framingham, Massachusetts, became a cohort of people followed forward in time in the Framingham Heart Study. This long-term cohort study has contributed to much of the knowledge we have today about cardiovascular disease and hypertension.[8] Information we use in clinical practice of the effects of diet, exercise, and medications (particularly aspirin use) were based on the results of this study. The cohort is now in its third generation of participants[9] contributing to study results that now include genetic information. Physical health examinations ended for the original cohort in 2014.

Summary Comparison Tables of Study Designs

Table 14.2 is a summary of the study designs used in pharmacoepidemiology. Table 14.3 is a summary of the advantages and disadvantages of each study design and results given.

Data Used in Pharmacoepidemiological Research

Two types of data are used in pharmacoepidemiological research: primary and secondary data. The type of data appropriate for the study depends on the research question and the study design. Primary data are collected for a specific purpose and

TABLE 14.2 Summary Table of Studies Used in Pharmacoepidemiology

Study Type	Groups	Measure of Association	Purpose
Ecological/cross-sectional study	Sample of population	Prevalence	Assess prevalence Describe population characteristics used in health planning
Case–control study	Cases vs. controls	Odds ratio	Association between disease and historical exposure Useful rare diseases Relatively easy
Cohort study	Exposed vs. unexposed	Relative risk	Association between a risk factor and future outcome Useful rare exposures Expensive

TABLE 14.3 Advantages and Disadvantages for Each Study Design

Study Type	Advantages	Disadvantages
Ecological/cross-sectional study	• Rapid, easy, and cheap • Provides evidence for or against a hypothesis • Can be used in international comparisons	• Only study groups and not individuals • Cannot control confounders • Cannot differentiate cause and confounders • Cannot identify nonlinear relationships
Case–control study	• Identifies rare outcomes • Evaluates multiple risk factors and exposures • Used in diseases with long latent periods • Follow-up is not a problem • Relatively fast, easy, and cheap	• Selection of controls could lead to selection bias • Controls should represent the general population • Retrospective data on exposure may lead to recall bias
Cohort study	• Can measure the incidence and prevalence of disease • Can look at multiple outcomes in a study • Provides a clear temporal association	• Needs a large sample size to look at rare outcomes • Expensive • Long-term commitment of the population to the study • Chance of loss to follow-up

have not been previously used in research. These data can be collected through questionnaires, interviews, focus groups, or medical note extractions. Primary data differ from secondary data in that they offer the researcher increased control over what type of and how much data are collected. For example, if a researcher wanted to find out if patients take their tablets with a meal or not, this question should be asked. These are primary data; they would not be found in any other database or resource. The main disadvantages in collecting primary data are time and cost. Interviews, chart reviews, and questionnaires take time to prepare and analyze. As the sample size of the study increases, the cost of additional people to review notes also increases, which is an additional cost.

Secondary data are preexisting and were collected for another purpose. These may include medical administrative data (e.g., hospital admissions), prescription dispensing data, or data from randomized controlled trials. These types of data have the distinct advantage of being available to the research without waiting for the data to be collected. Secondary data are usually large in sample size and offer generalizability.[10] These strengths have resulted in secondary data being used in a variety of research topics, including drugs utilization, adherence to medication therapy, healthcare interventions, and policy issues. Increasingly, secondary data are becoming more available with computerization of medical records and billing procedures. There are some disadvantages though. Secondary data, however, are usually not collected for the purposes of research, but for other intentions (e.g., billing customers, record keeping, prescription filing) and therefore may not have all the information that is required to complete the study. For example, clinical lab data may not be available at the individual level. With advanced statistical techniques and data management (linking dataset by a unique identifier), many of these disadvantages can be overcome.

Postmarket Surveillance

Postmarketing surveillance (PMS) is defined as the identification and collection of information regarding drugs after their approval for use in a population.[11] The drug approval process in some countries is complicated and lengthy, which may hold drugs back from patients in desperate need of them. PMS is a method of systematically monitoring the safety and effectiveness of new drugs in the real world using a variety of patient types with many different comorbid diseases. The population of potential users after a drug is released is very different from the population studied in the premarking phase of a drug's approval. For example, few clinical trials will include very old patients or patients with two or more comorbidities or women that are breastfeeding.[11]

PMS allows for the long-term monitoring of the effects of drugs. This contrasts the follow-up period of randomized controlled trials that are usually shorter in duration when considering the cost of the trial.[11] The long-term effects, such as tolerance to the drug or adverse drug reactions, use PMS study designs. Especially rare adverse events that may not be identified in clinical trials because of the small sample side, PMS data may include thousands of patients using the medication over a period of time allowing for these rare events to be quantified and studied.

Case Study: Postmarketing Surveillance Study

Postmarketing surveillance study of the safety and efficacy of sildenafil prescribed in primary care to erectile dysfunction patients.[18]

The safety and efficacy of sildenafil use in primary care were studied in a total of 651 men with erectile dysfunction enrolled from primary care in Korea. Patients were followed up, and all adverse drug reactions and efficacy data were collected. 458 patients completed the study. The study found hypertension and diabetes were associated with poor efficacy. A total of 71 adverse events were reported in 56 patients (8.6%). The study concluded that sildenafil prescribed by primary care physicians was well tolerated and improved erectile function in patients with erectile dysfunction.

PMS also allows for other indications for medication use to be observed and evaluated. "Off-label" use or using a medication for another indication not included in the official drug information can be identified and evaluated. Therefore, knowledge gained from PMS allows for the broader application of drugs to special populations, for different indications, and at doses and durations not studied in the prelaunch clinical trials.[11]

Pharmaceutical Policy Issues

PMS can be used to ensure that appropriate and effective pharmaceutical policies are developed to ensure fair and equitable use of drugs in a population. Data on how and where a drug is used after its launch into a large population can inform decision-makers and drug regulators on innovation and service provision, issues of access to drugs, pricing and containment, and rational use of drugs. With additional public health data, such as socioeconomic data, age and sex distributions, and hospital admissions, additional issues of unmet need can be identified and quantified for use at a larger population level.

PMS can provide information about drug safety and effectiveness of drugs that can be used by the public, manufacturers, and government to make informed choices. Rare adverse events from long-term use and in large populations may inform the safe use of drugs. In addition, the effects of nonadherence to therapy can be identified. PMS will continue to be an important "phase" of the clinical approval process of drugs introduction to the market.

OBJECTIVE 14.3. DISCUSS HOW PHARMACOEPIDEMIOLOGY INFLUENCES HEALTH IMPROVEMENT

Identifying the people who need the most pharmaceutical care in a community is a difficult undertaking, but with the right data, services and products can be targeted to the people who need it the most.[12] Developing and producing a pharmaceutical needs assessment will identify those most in need. This may not be what the community wants, but "wants" are based on other factors too. Pharmaceutical needs assessments will direct the services provided to the areas that will benefit most. For example, smoking cessation services to areas of high incidence of asthma and COPD or flu vaccination clinics for the elderly in areas where there is a large aging population. This ensures necessary public health services will be provided where they will maximize the benefit to the most people.

The natural history of a disease can have many areas that might need pharmaceutical interventions to prevent and promote healthy states.[13] Pharmacists can intervene with a patient who is at risk of a disease by providing a smoking cessation program to prevent future COPD or flu vaccinations to prevent the disease, especially in vulnerable populations. This will maintain the health of the population. However, if the intervention is lacking, the disease will progress and may require advanced clinical pharmacy services in a hospital environment during the illness to the point of recovery or death depending on the severity of disease and the patient's circumstances. Therefore, the points of intervention in the natural history of the disease may require unique skills and competencies gained by the pharmacist.

Pharmacoepidemiology by its innate nature assesses and influences decisions in public health. Public health is defined as a discipline concerning itself with improving health or preventing illness in a population, and it is usually implemented by a government or a group accountable to a community.[14] Public health practitioners use data that are readily available, including population health surveys, census data, data on educational attainment, crime and police data, and other sources to determine the health of the population. Pharmacoepidemiology can provide additional information about a population that would not normally be found in the public health data. Pharmacoepidemiology can be used to identify and describe patterns of use or adverse drug reactions, compare the actual use of a medication to the guidelines or expected use patterns, determine factors that promote or inhibit the use of medications, and link usage to outcomes. Pharmacoepidemiology can use postmarket surveillance data to identify medication safety issues and patterns of use in a population.[14] Therefore, the end result will be several sources of information on a population so that the practitioner can make an informed, rational choice for the population.

Vaccination Services and Disease Prevention: an Example of Pharmaceutical Public Health

A widespread outbreak of H1N1 influenza virus in 2009 once again initiated the debate of vaccination of children and adults against the flu virus. Questions like "is it really necessary?" or "isn't it like a bad cold?" were asked by patients without enough information to make an informed decision. Pharmacists became instrumental in providing patients with pertinent information to make informed choices by talking about the risks and benefits of immunization against the virus. Pharmacists may also ease the fears of many patients by providing the facts as well as the significant risks associated with not being vaccinated against flu. Pharmacists are in a unique position to identify those patients that are at most risk of infection, whether the patient is in a hospital or the community pharmacy, ensuring the right patients are targeted for vaccination.[15]

As of 2017, globally, more than 20 countries allow vaccinations in a pharmacy or by a pharmacist. Of these, 13 allow pharmacists to vaccinate independently. This demonstrates the expansion and growing acceptance of pharmacy immunization services around the world.[16] In the United States (USA), all 50 states allow pharmacists to administer vaccinations with appropriate training. Immunization by pharmacists is governed by each state's laws and regulations governing pharmacy practice. Some states require specific education or certification; some limit the types of vaccinations given by pharmacists.[15] Nevertheless, the patient can now make their own informed choice to get a vaccination by a knowledgeable healthcare professional at their convenience. Many community pharmacies are open long hours and provide necessary public health services.

Case Study: Pharmacists Vaccinate Against Cancer

Lowery M. Pharmacists key to widespread HPV vaccination.[19]

Millions of avoidable cancers could be prevented through widespread administration of the human papillomavirus (HPV) vaccination. In the USA, states were encouraged to enact laws allowing pharmacists to administer the HPV vaccination. The reasons behind this recommendation included reducing missed clinical opportunities to recommend and administer the vaccines; increasing parents' and adolescents' acceptance of the vaccines; and maximizing access to HPV vaccination services. Presently, about one-third of girls aged 13–17 years (prevention) and 7% of boys 13–17 years (transmission) have been vaccinated. More young people need to be vaccinated to decrease the risk of cervical cancer even more in the future.

Decision Making

Some drugs are expensive, and new drugs arriving on the market have to be carefully controlled so that the resources of the healthcare system are not used in one particular area. There are innovative methods of treatment being developed that also require new resources, trained people, and money to buy-in the products. Decisions have to be made so that there is rationalizing (not rationing) of the budget to ensure that as many people as possible will benefit from new and existing drugs and procedures. Using data from many sources, including clinical practice and from the literature on the effectiveness of treatment, provides the evidence needed to make informed decisions. With this in mind, it is possible that not everyone will benefit. There are those that have not tried all other treatments and failed or would like a quick fix. However, there may be a large group that requires a moderately priced drug to maintain the status quo of the disease. Guideline development and drug formularies are guidance for practitioners for best practice based on the available evidence that is presented in the literature. Nevertheless, the practice of drug still requires clinical judgment to ensure the right drugs are provided to the right patient safely and effectively.

The people making the decisions realize that they require the evidence to "back up" their decisions. Robust pharmacoepidemiological data, strong study designs, and appropriate use of statistics will give the necessary evidence to ensure that use of the drug is safe and the drug is used in situations where there are positive outcomes. Decision making still needs an understanding of the one variable in this equation—the patient. Knowing the effects of the determinants of health on a patient, other factors must also be considered, which are not always recorded in a database. Communication and analyzing the conversation with the patient will provide additional data to ensure the patient is right for the treatment of choice.

Determinants of Health

When determining the health of a population, it is important to consider the conditions and factors associated with health. These include the wider perspective of living conditions as well as the health-related factors within the individual. The factors are viewed in levels: the individual fixed conditions, lifestyle factors, the social and community level, and finally general socioeconomic, cultural, and environmental conditions (Fig. 14.6).[17]

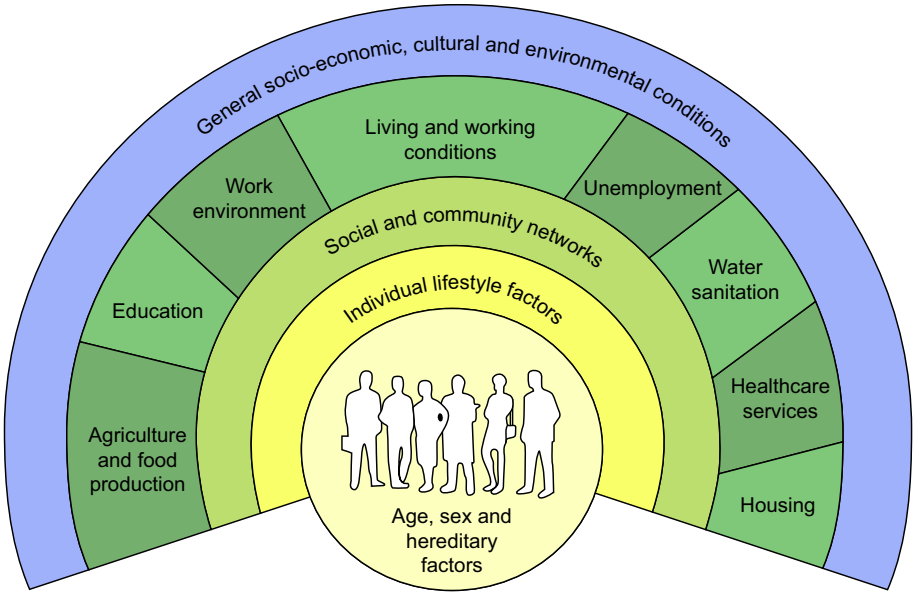


FIGURE 14.6 The determinants of health and the relationship to the individual. *Adapted from Dahlgren, Whitehead. Policies and Strategies to Promote Social Equity in Health. Stockholm, Sweden: Institute for Future Studies; 1991. [Reprinted in 2007].*

For individuals (center of diagram) there are fixed conditions that cannot change and can determine the health of an individual. These include age, sex, and genetics. The next layer includes lifestyle factors that can be changed or modified to prevent or promote health in an individual. These are primarily behavior changes that an individual makes, including smoking cessation, decreasing alcohol consumption, and wearing a seat belt while driving. The next layer contains social and community influences that provide support from friends and family when in unfavorable conditions. The final layer is structural factors that influence health. This layer is not intuitive; it does make sense how housing could affect health. For example, if a housing area has many drafty, damp, and moldy houses, it increases the chances of developing asthma by nature of the environment. Pharmacoepidemiological data are used in this example to evaluate the use of asthma medication in this area and test the hypothesis compared with another area of lesser risk.[17]

CONCLUSION

Pharmacoepidemiology is an emerging field to understand the use of medications in a large population, identify and quantify adverse drug reactions in a population, and to quantify the risk or benefit of taking a medication for a particular disease or condition. Increasingly, secondary data are becoming more available with computerization of medical records and billing procedures. Pharmacoepidemiology has established and robust study designs that provide strong evidence for the drugs in the relationship to outcomes. Pharmacoepidemiology results can be used in conjunction with other data to explain or identify areas of need and improve the health outcomes.